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PROJECT MANAGEMENT.

Projects are a unique aspect of business operations that require a special management approach.

Many other aspects of business tend to operate more routinely whereas projects have uncertainties and risks that tend to make them more challenging.

Examples of projects are

(i) Hosting Olympic Games } Very huge
(ii) " World Cups }
(iii) Producing a Motion picture
(iv) Producing a segment of a popular television offering e.g. "American Idol" } Small in

They involve a tremendous amount of

(i) planning
(ii) Preparation
(iii) Coordinating work

All these need to get done before and during the ~~games~~ entire project period/d

Not all projects are successful, and the consequences of project failure can be costl and even catastrophic.

Many projects are going on all around us. everyday — some are large and some are Project Management is important in ensuring t

certain kinds of work get done.

WHAT IS A PROJECT??

Projects are unique, one-time operations designed to accomplish a specific set objectives in a limited time frame.

Other examples are:

(i) Construction of a shopping complex
(ii) merging two companies
(iii) Designing new products
(iv) reengineering a process
(v) Developing Web pages.

Projects may involve considerable cost. Some have a long time horizon, and some involve a large number of activities that must be carefully planned and coordinated.

Most are expected to be completed based on time, cost and performance targets.

To accomplish this, goals must be estsblis and priorities set.

Tasks must be identified and time estimate ~~set~~ made

Resource requirements must be projected and budgets prepared.

Once under way, progress must be monitored to assure that project goals and objectives will b achieved.

AN OVERVIEW OF PROJECT MANAGEMENT.

WHAT IS PROJECT MANAGEMENT??

A team-based approach for managing projects

HOW IS IT DIFFERENT FROM GENERAL OPERATIONS MANAGER

(i) Limited time frame
(ii) Narrow focus, specific objectives.
(iii) Less bureaucratic.

WHY IS IT USED.

(i) Special needs that don't lend themselves to functional management.
(ii) Pressures for new or improved products or services, cost reduction.

WHAT ARE THE KEY ~~SUCCESS FACTORS~~ METRICS

(i) Time
(ii) Cost
(iii) Performance objectives

WHAT ARE THE KEY SUCCESS FACTORS??

(i) Top-down commitment
(ii) A respected and capable project manager
(iii) Enough time to plan
(iv) Careful tracking and control
(v) Good communications.

HAT ARE THE MAJOR ADMINISTRATIVE ISSUES

1. Executive responsibilities
   (a) Project selection
   (b) Selecting of a project manager
   (c) organisational structure (To whom will the project manager report).

2. Organisational Alternatives:
   (a) Manage within functional unit
   (b) Assign a coordinator
   (c) Use a matrix organisation with a project leader.

WHAT ARE THE MAIN TOOLS??

1. Work Breakdown structure: An initial planning tool that is needed to develop a list of activities sequences and a realistic budget.
2. Network Diagram — A "big picture" visual aid that is used to estimate project duration, identify activities that are critical for tainely project completion, identify areas where slack time exists, and develop activity schedules.
3. Gantt charts — A visual aid used to plan and monitor individual activities
4. Risk Management: Analysis of potential failures or problems, assessment of their likelihood and consequences and contigency plans.

KEY DECISIONS IN PROJECT MANAGEMENT.

Much of the success of projects depend on

(1) Deciding which projects to implement
This involves determining the criteria that will be used. Factors include budget, availability of appropriate knowledge and skill personnel, cost-benefit considerations.

(2) Selecting the Project Manager
This is the central person in the project.
He is responsible for managing:
(i) the work
(ii) Human resources — those working have direction and motivation
(iii) Communications. everybody has the information needed to do the work.
(iv) Quality
(v) Time — project is completed on schedule
(vi) Costs — completed within budget.

The position of Project manager has high visibility.

(3) Selecting the Project Team — how well the person works with others.

(4) Planning and Designing the Project — ~~Lecturer~~
Requires decisions on project performance goals.

timetable for project completion, the scope of the project, what work needs to be done, how it will be done, if some portions will be outsourced, a budget and when and how long resources will be needed.

5. Managing and Controlling Project Resources — involves managing personnel, equipment and the budget

6. Deciding if and when a Project should be terminated — important considerations are likelihood of success, termination costs and whether resources could be better used elsewhere

PROJECT LIFE CYCLE.

Projects go through a life cycle which typically consists of four phases:

1. Definition.
   (a) Concept — the organisation recognises the need for a project, responds to a request for a proposal from a client.
   (b) Feasibility analysis — examines the

expected costs, benefits and risks of undertaking the project

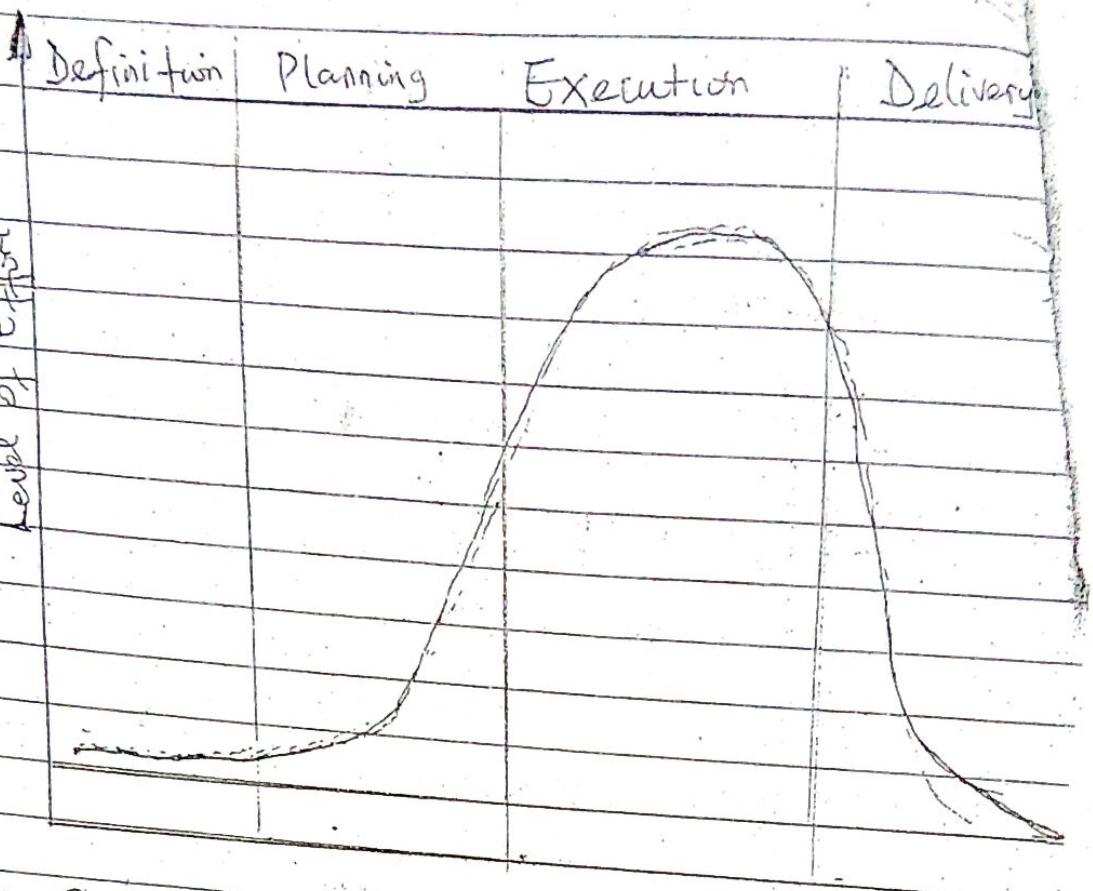
2. PLANNING :- This spells out the details of the work and provides estimates of the necessary human resources, time and cost.

3. EXECUTION — Here, the project is done. This phase accounts for the majority of time and resources consumed by a project.

4. TERMINATION :- Closure is achieved. Deals with leftover materials, equipment and any other resources associated with the project.

It should be noted that phases can overlap. This can reduce the time necessary to move through the life cycle and therefore generating cost saving.

• PROJECT LIFE CYCLE.



1. Goals	1. Schedules	1. Status reports	1. Train customers
2. Specifications	2. Budgets	2. Changes	2. Transfer document
3. Feasibility	3. Resources	3. Quality	3. Release resources
4. Tasks	4. Risks		4. Reassign staff
5. Responsibilities	5. Staffing		5. Lessons learned
6. Teams			

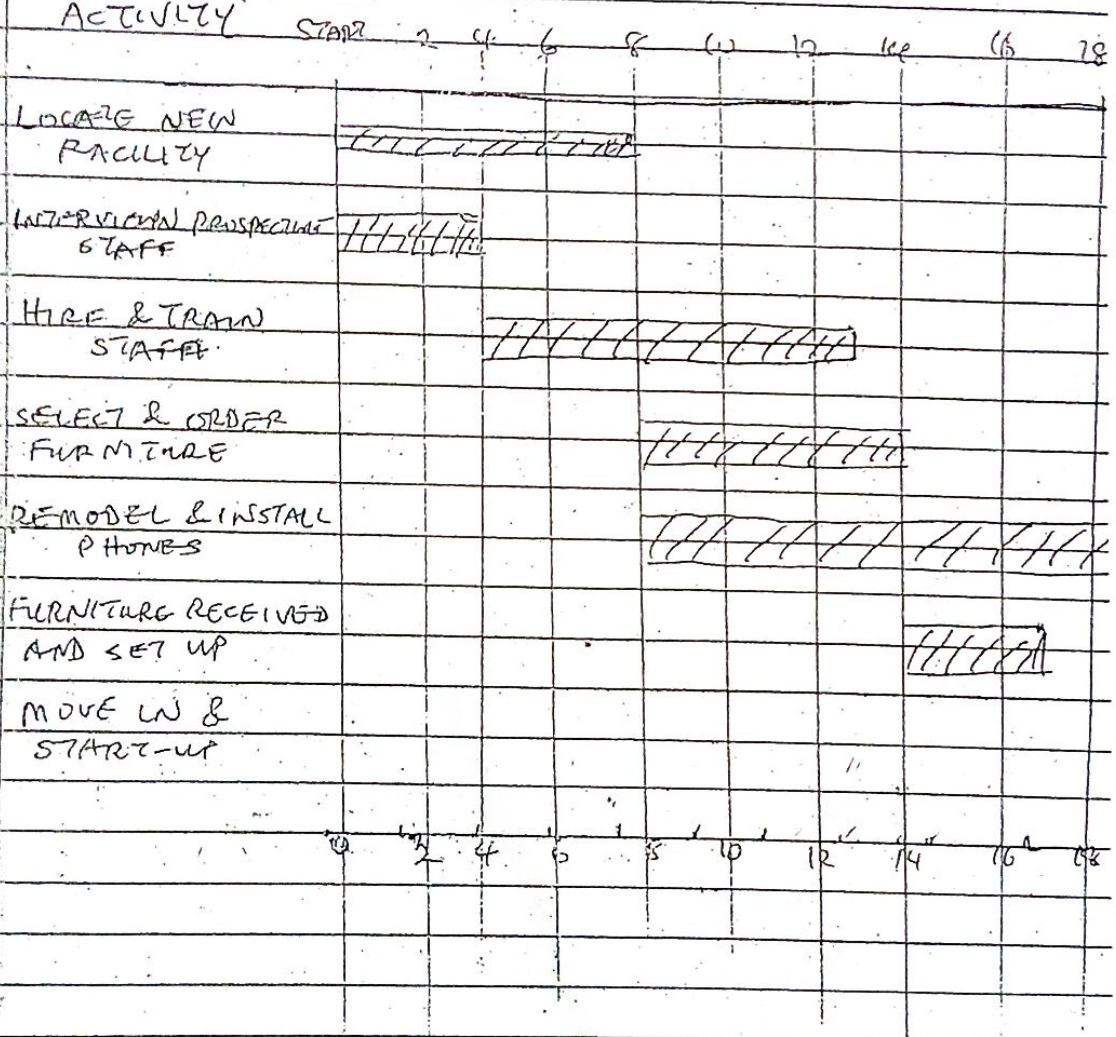
PLANNING AND SCHEDULING WITH GANTT CHARTS.

The Gantt chart is a popular tool for planning and scheduling simple projects. It enables a manager to initially schedule project activities and then to monitor progress over time by comparing planned progress to actual progress.

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- Process Re-engineering
- Human Capital Development
- Specialized Training

Its advantage is that it is simple.



PERT AND CPM.

PERT (Program Evaluation and Review Technique) and CPM (Critical Path method) are the two most widely used techniques for planning and coordinating large-scale pr

- ☐ Strategy & Business Support
- ☐ Process Re-engineering
- ☐ Human Capital Development
- ☐ Specialized Training

By using these 2 techniques, managers are able to obtain

(1) A graphical display of project activities
(2) An estimate of how long the project will take.
(3) An indication of which activities are the most critical to timely project completion.
(4) An ~~idea~~ indication of how long any activity can be delayed without delaying the project.

NETWORK DIAGRAM
Diagram of project activities that shows sequential relationships by use of arrows and nodes.

ACTIVITY-ON-ARROW (AOA)
Network diagram convention in which arrows designate activities

ACTIVITY-ON-NODE (AON)
Network diagram convention in which nodes designate activities

PATH
A sequence of activities that leads from the

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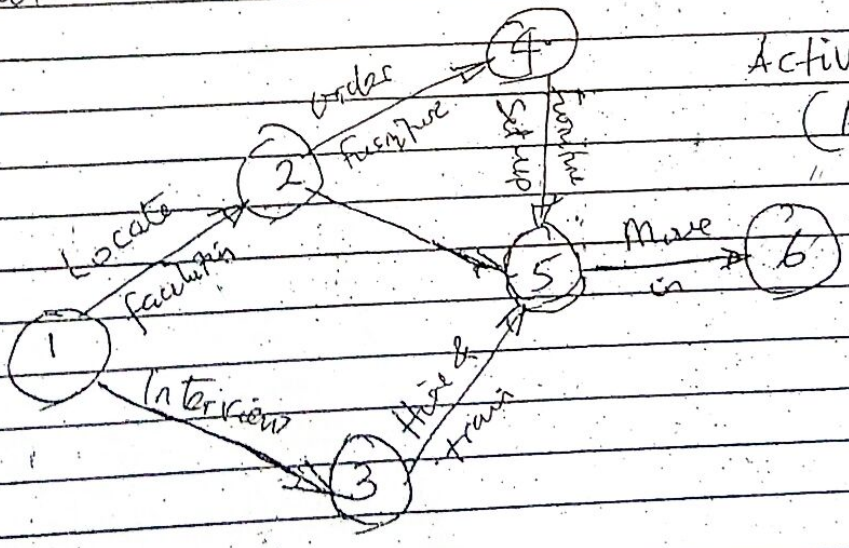
Strategy & Business Support
Process Re-engineering
Human Capital Development
Specialized Training

starting node to the finishing node.

CRITICAL PATH
The longest path, determines expected project duration.

CRITICAL ACTIVITIES
Activities on the critical path.

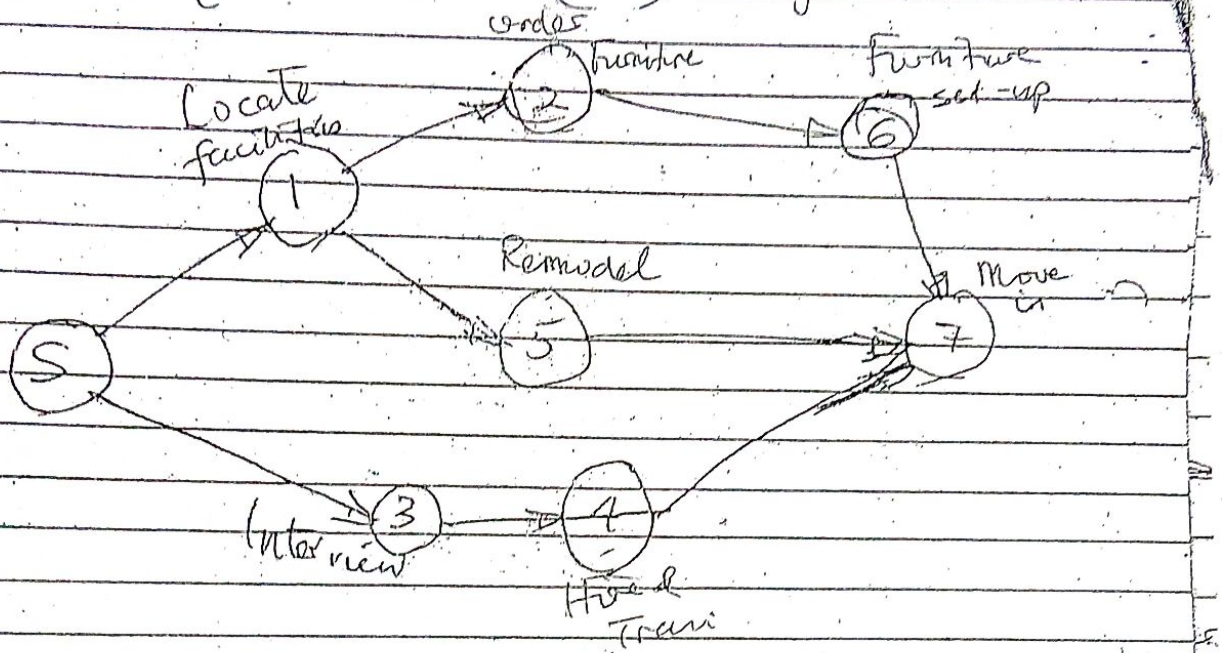
SLACK
Allowable slippages for a path; the difference between the length of a path and the length of the critical path.



Activity-on-Arr
(AOA) diagr

- Strategy & Business Support
- Process Re-engineering
- Human Capital Development
- Specialized Training

Activity - on - Node (AON) diagram



EXAMPLE.

Given the additional information on the bank network below, determine

(a) The length of each path.

(b) The critical path

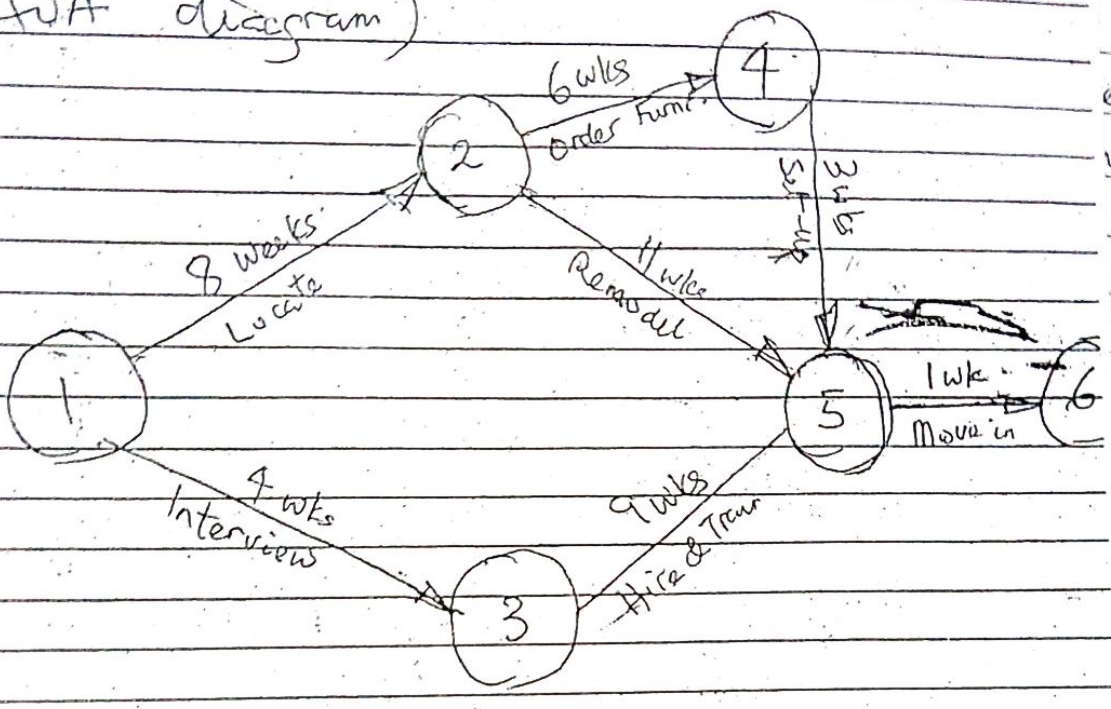
(c) The expected length of the project

(d) The amount of slack time for each paths

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(AOA diagram)



(a) (i) Path 1-2-4-5-6 = 8+6+3+1
= 18 wks.

Path 1-2-5-6 = 8+11+1 = 20 wk

Path 1-3-5-6 = 4+9+1 = 14 wk

(b) Path 1-2-5-6 is the longest path (20 wks), So it is the critical p

(c) The expected length of the proje is equal to the length of the critical path (i.e; 20 wks)

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(d) We find the slacks for each path by subtracting its length from the length of the critical path.

Path	Length (wks)	Slack (wk)
1-2-4-5-6	8+6+3+1 = 18	20-18 = 2
1-2-5-6	8+11+1 = 20	20-20 = 0
1-3-5-6	4+9+1 = 14	20-14 = 6